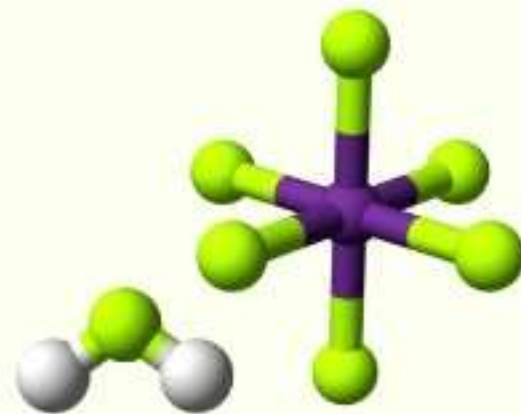


CHEMISTRY

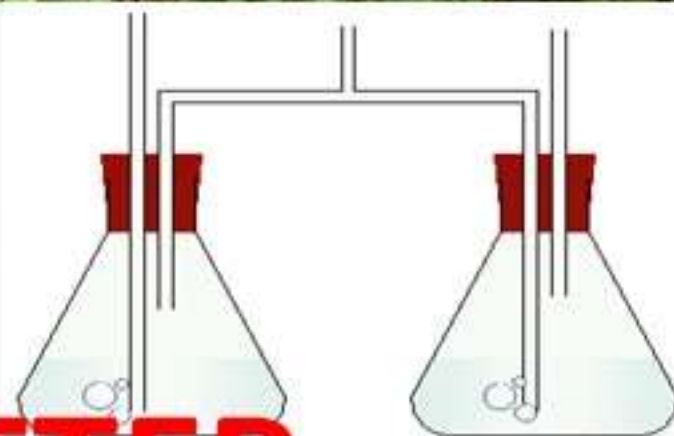
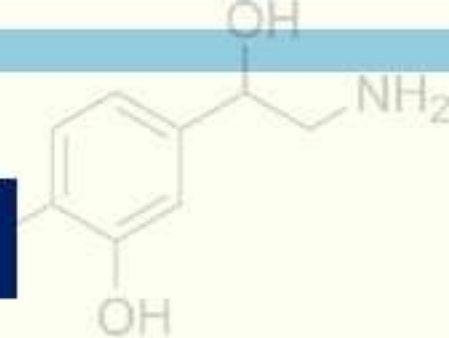
THE WORLD OF MATTER - 1

MODULE - 1

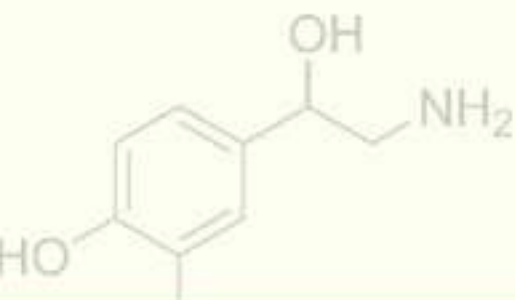


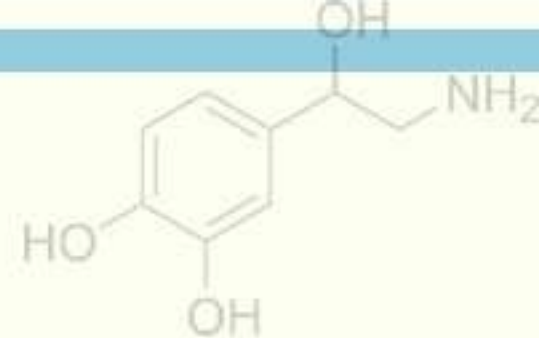


WHAT IS EVERYTHING MADE UP OF?



MATTER



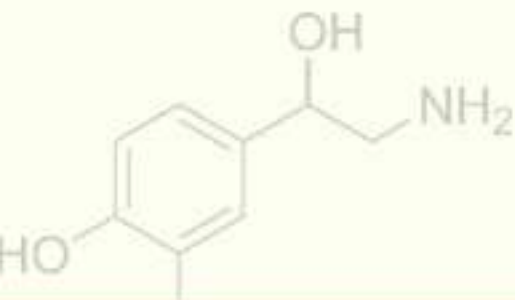
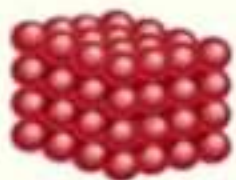


WHAT IS MATTER?

MATTER IS ANYTHING THAT HAS MASS AND OCCUPIES SPACE

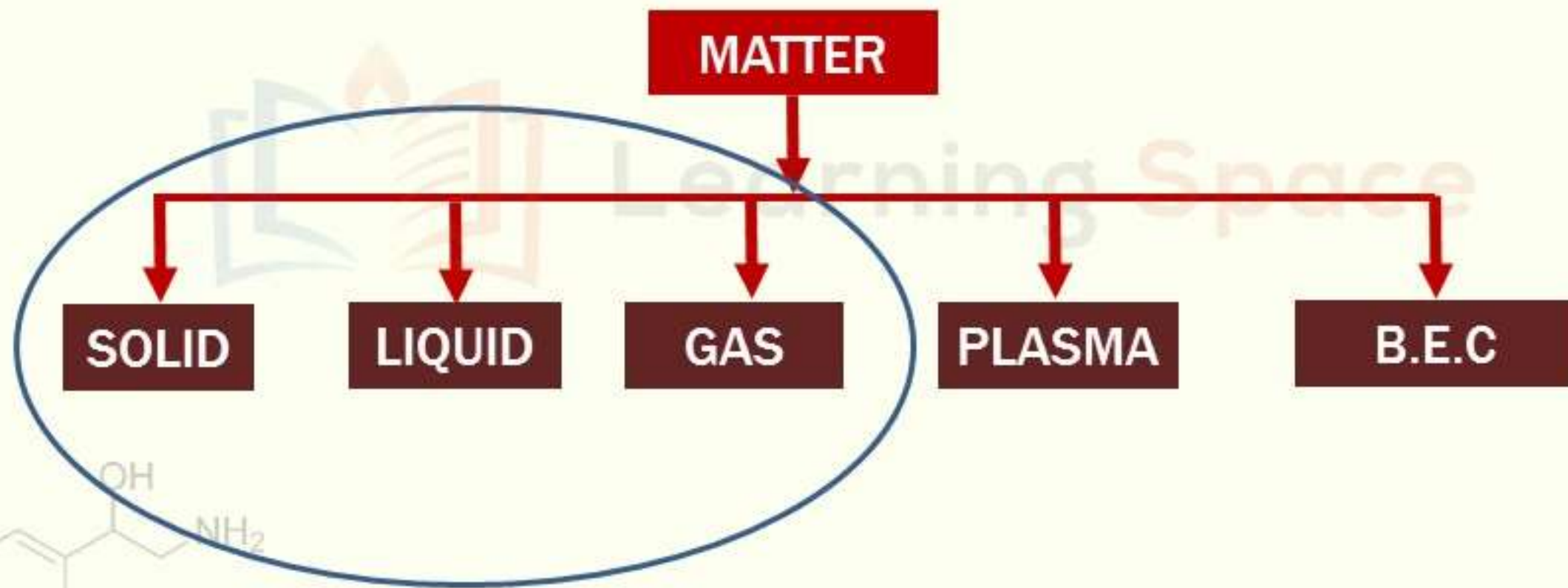


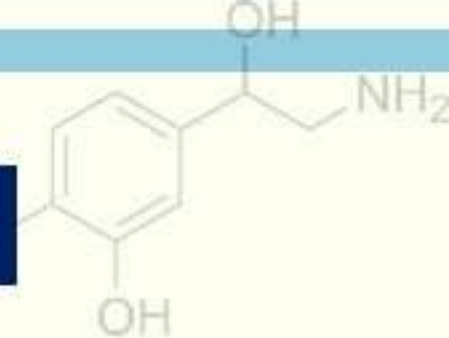
Learning Space





MATTER EXIST IN DIFFERENT FORMS





MATTER EXIST IN DIFFERENT FORMS

THE KNOWN STATES OF MATTER

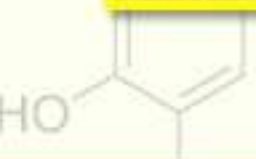
1. SOLID



2. LIQUID



3. GAS





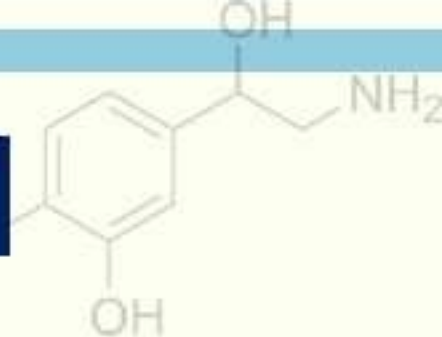
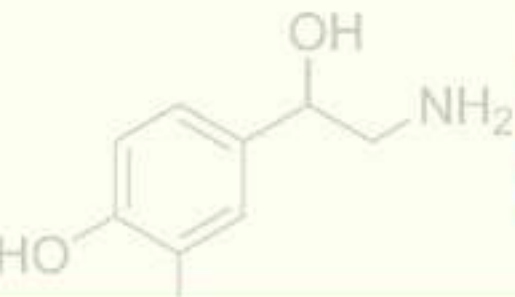
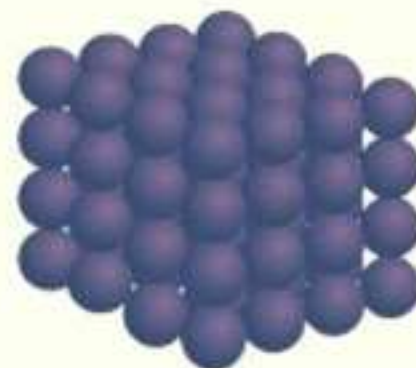
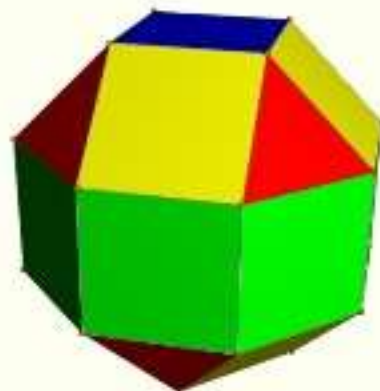
MATTER EXIST IN DIFFERENT FORMS

THE STATES OF MATTER

1

SOLID

- Solid particles are tightly packed has a definite shape and definite volume
- They are not compressible, cannot be further compressed in more tight shape
- They show the property of Elasticity
- some solid can retain their shape after an extended force is withdrawn like an elastic rope or a rubber balloon





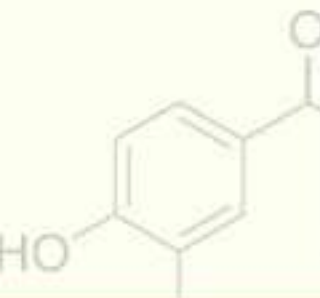
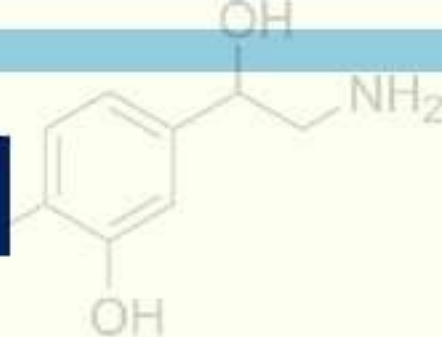
MATTER EXIST IN DIFFERENT FORMS

THE STATES OF MATTER

2

LIQUID

- Liquid particles are packed less tightly than solid so that is the reason they can flow.
- Liquids and Gases are thus called Fluids.
- They are non compressible and has a definite volume. They don't have a definite shape.





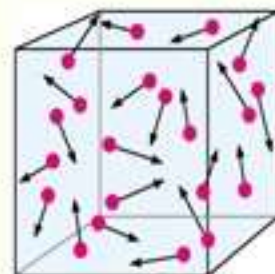
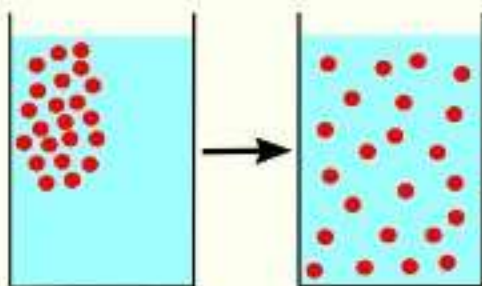
MATTER EXIST IN DIFFERENT FORMS

THE STATES OF MATTER

3

GAS

- Gas particles are very sparse from each other
- They neither have a definite shape nor a definite volume.
- The gas particles also can flow very fast and move in a random motion(**Brownian** motion).
- They can flow from their concentrated areas to other areas (**diffusion**)
- They can be highly compressible and can be stored under pressure(LPG cylinder, Coca Cola bottles)





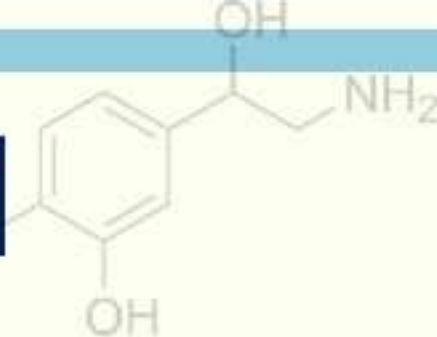
MATTER EXIST IN DIFFERENT FORMS

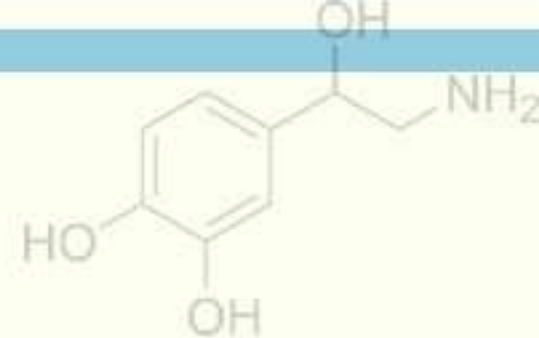
THE STATES OF MATTER

4

PLASMA

- Plasma forms when **gases** are heated at such **high temperatures** that they start glowing
- Sun is a plasma. **The Hydrogen gas gets converted to Helium**. This reaction generates tremendous amount of heat that the gases glow and emit huge amounts of light.
- Other examples of plasma includes lightning, bright neon signs and tube lights, the ionosphere, the tail of comet etc.





MORE EXAMPLES OF PLASMA

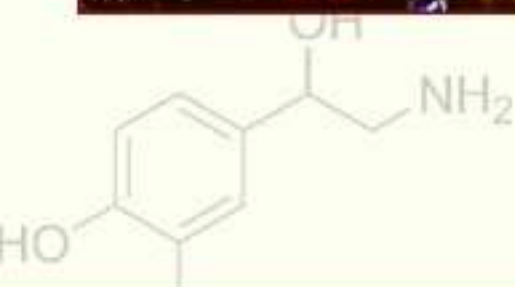
NEON SIGN BOARDS



FLORESCENT TUBES



COMET'S TAIL





MORE EXAMPLES OF PLASMA

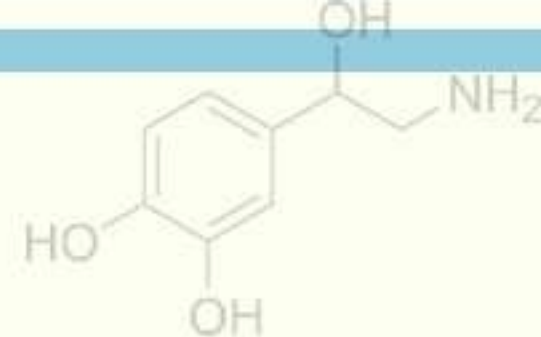
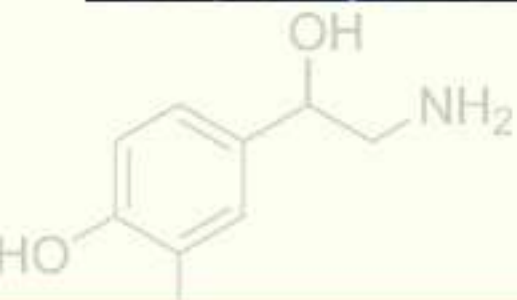
STARS



IONOSPHERE

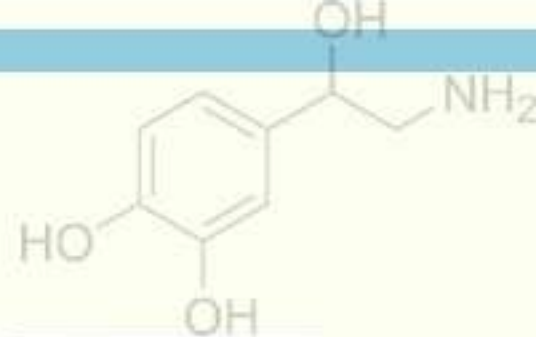


LIGHTNING



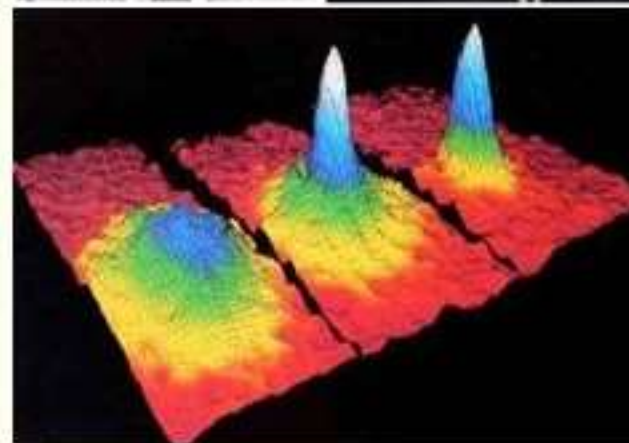
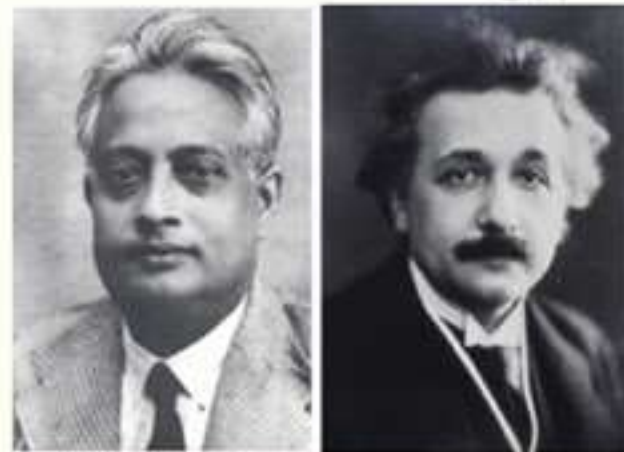


BOSE EINSTEIN CONDENSATE



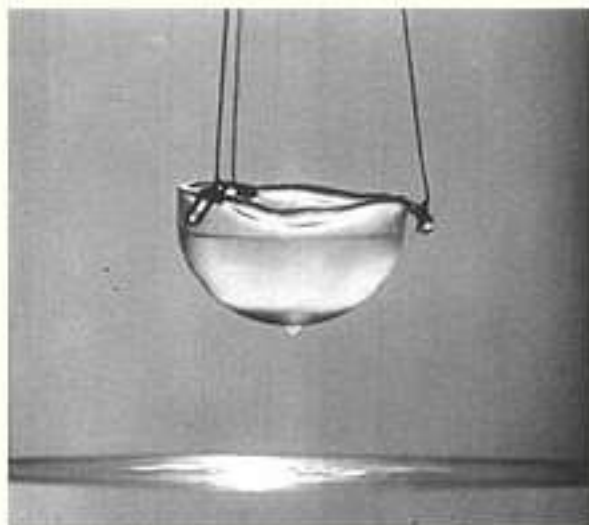
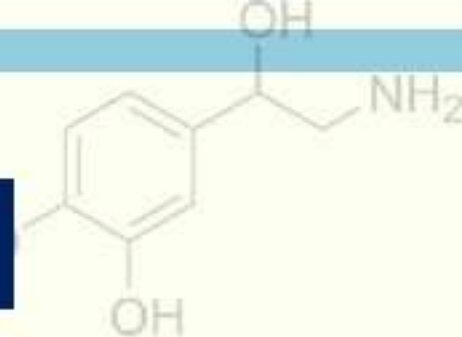
1 Bose Einstein Condensate is also called the fifth state of matter
A BEC is formed by **cooling a gas of extremely low density**, about $1/100000$ the density of normal air, to **ultra-low temperatures** (conceptual).

2 These dilute gas are cooled to their absolute zero and are called as **Bosons**; Satyendra Bose and Albert Einstein predicted the existence of such a state in the 1920s, but it was not observed until 1995 by Eric Cornell and Carl Wieman.



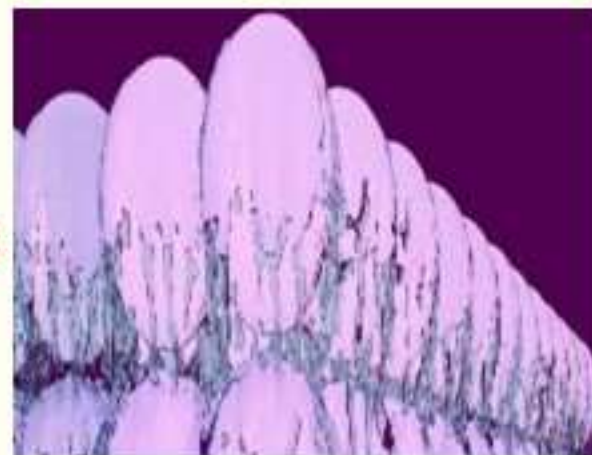


STILL MORE STATES OF MATTER !!!

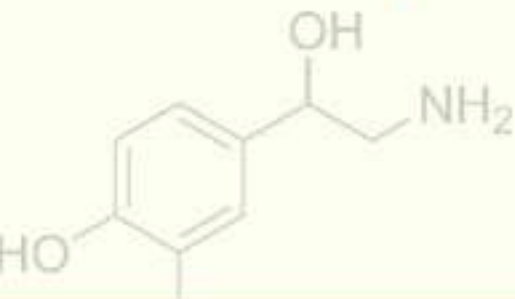


superfluids

Scientists have found **superfluids** and **supersolids**, which are **cryogenic** liquids and solids that are able to flow or move **without friction** at extremely low temperatures.

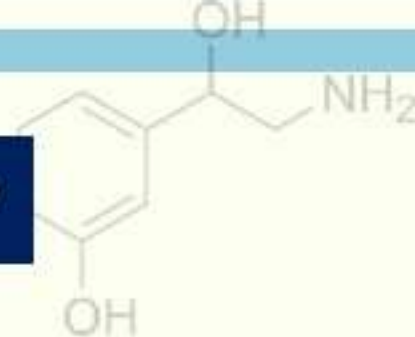


supersolids





HOW DO WE SEE MATTER IN OUR DAILY LIFE ?

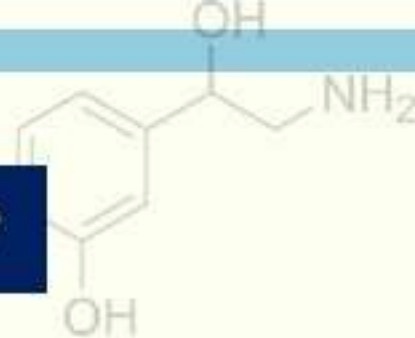


Matter

1. Pure
Matter

2. Impure
Matter (Mixture)





HOW DO WE SEE MATTER IN OUR DAILY LIFE?

1. PURE MATTER

- 1 Pure Matter are those substances that have only one type of particles in them, or they behave as a single unit.
- 2 They have a **definite and constant composition**.
- 3 The constituents cannot be separated by **physical means**.





HOW DO WE SEE MATTER IN OUR DAILY LIFE?

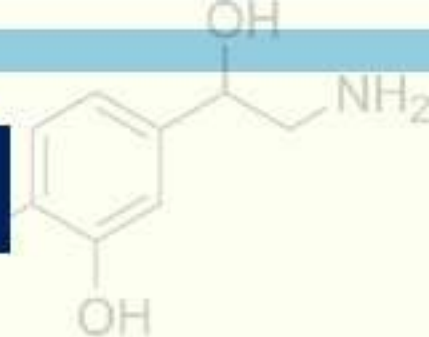
ELEMENTS**1. PURE MATTER****COMPOUNDS**

- Elements are those substances which is made up of only one type of atoms inside them.
- Eg. Copper metal is made up of only one type of atoms of copper.



- Compounds are the substances which is made up of two or more elements who are bonded chemically.
- Eg. Water made up of 3 atoms, 2 Hydrogen & 1 Oxygen atom.





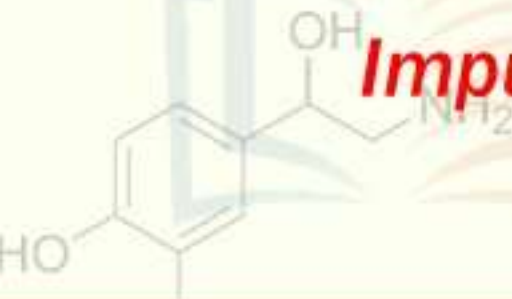
HOW DO WE SEE MATTER IN OUR DAILY LIFE?

LET'S SEE WHAT WE FIND IN A COMMON MIXTURE ...

- 1 Mixtures are a combination of two or more elements or compounds
- 2 They can be separated by simple physical means.
This means the constituents retain their original characters even after mixing with others.

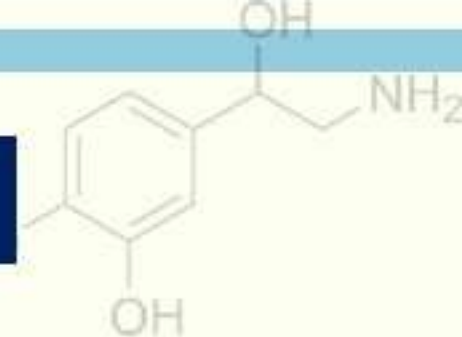


Impure matter are called mixtures and are classified into many types!





CLASSIFICATION OF IMPURE MATTER



Impure Matter

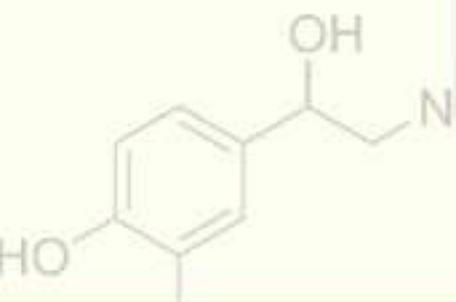
Homogeneous
mixture

Heterogeneous
mixture

solution

suspension

colloid



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0 8 6 6 -2 4 4 4 4 7 2
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